

access data from the user's contacts to set up an appointment or even from the notes application.

Earlier versions of the OS fared poorly with regards to handwriting recognition, specifically cursive handwriting. This remained one of Newton's drawbacks, and many remarked that the absence of this feature alone could have helped the MessagePad become a greater success. Future iterations of the device improved upon handwriting considerably, and it could identify text written in block letters (a system called 'Rosetta') very well. Indeed, this technology remained unrivalled upto 10 years after production ended. Several other gesture-based interactions such as the ability to 'strike-out' a wrongly entered word or to encircle a word to 'highlight' it were built in, and the OS sported 'in-situ' word correction, where a user simply had to double tap a wrongly interpreted word and a list of alternatives would pop down, letting the user choose the correct one. Text could also be entered via a virtual keyboard that would pop up when required. So strong was the support for pen-based gestures, that the MessagePad would identify when the user was trying to draw a shape such as a line, circle or a polygon, and would correct the shape into an exact vector figure. These 'Shapes' would automatically be assigned specific vector points (such as vertices for a polygon) which could be changed manually. Also supported was 'Ink Text' which was a form of searchable free-hand text. Additional formatting options like word-wrapping and bold, italics and underline were also supported for 'Ink Text'. Thus a note saved using the Notes application could have any combination of free-hand sketches, shapes, handwritten text or printed text entered via the virtual keyboard.

In terms of connectivity, Apple included highly advanced features such as Infrared radios and support for native modem and ethernet connectivity thanks to the included PC Card slot. Developers from the Newton community also wrote drivers for the IEEE 802.11 and Bluetooth protocols. By the use of peripheral adapters, auxiliary tasks such as printing and sending faxes was also made possible.

Inspite of having features so far ahead of its time, the MessagePads failed to generate the kind of hysteria generally associated with Apple devices today. From a technical perspective, the Newton had a memory fragmentation issue which required users to reset the device every few months or so. The most frequently used application on the device, the notes application was notorious for being a memory hog. The device dimensions were uncomfortably large; for it could neither be operated with one hand (like a modern day cell-phone) nor could it be carried around in one's pocket, greatly reducing its mobility. The syncing features on the Newton were spotty at best, and the MessagePad soon turned